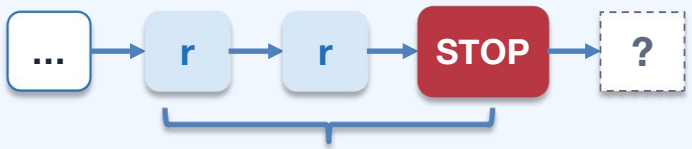
1 Termination Mismatch

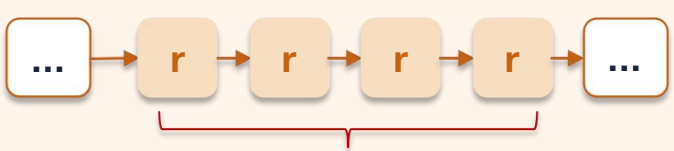
Premature Termination

Stops before reasoning is complete.



Repetition-Driven Over-Generation

Repeats after reasoning has stalled.



2 Targeted Objectives

Stop / Continue Distillation

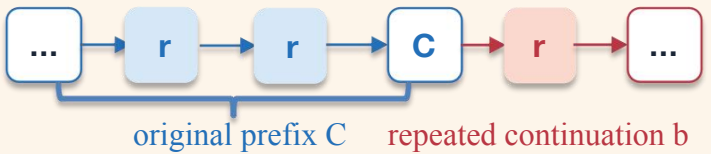
Align the continuation decision



$$\ell_{\text{per_token}} = q_{\text{EOS}} \log \frac{q_{\text{EOS}}}{p_{\text{EOS}}} + (1 - q_{\text{EOS}}) \log \frac{1 - q_{\text{EOS}}}{1 - p_{\text{EOS}}}$$

Repetition-Aware Unlikelihood

Penalize repeated continuations



$$\ell_{\text{per_token}} = -\log(1 - p(b_t \mid C, b_{<t}))$$

3 TA-OPSD

Combine the objectives to update the student.

Base OPSD objective

Full-vocabulary distillation

+

Stop / Continue Distillation

Align continuation decisions

+

Repetition-Aware Unlikelihood

Penalize repeated continuations

Update Student Model

Gradients update the student only